21-4a Do All Demand Curves Slope Downward?

Normally, when the price of a good rises, people buy less of it. This typical behavior, called the *law of demand*, is reflected in the downward slope of the demand curve.

As a matter of economic theory, however, demand curves can sometimes slope upward. In other words, consumers can sometimes violate the law of demand and buy *more* of a good when the price rises. To see how this can happen, consider Figure 13. In this example, the consumer buys two goods: meat and potatoes. Initially, the consumer's budget constraint is the line from point A to point B, and the optimum is point C. When the price of potatoes rises, the budget constraint shifts inward and is now the line from point A to point D. The optimum moves to point E. Notice that an increase in the price of potatoes leads the consumer to buy more potatoes.

Figure 13

A Giffen Good

In this example, when the price of potatoes rises, the consumer's optimum shifts from point C to point E. In this case, the consumer responds to a higher price of potatoes by buying less meat and more potatoes.

A graph, plotting quantity of meat on the horizontal axis and quantity of potatoes on the vertical axis, shows two downward sloping lines, both extending from point A on the horizontal axis to points B and D on the vertical axis, with point D closer to the origin. Line AB is the initial budget constraint line and line AD is the new budget constraint line; both are tangential to downward sloping indifference curves bowed in toward the origin, I1 and I2, at points marked C and E respectively. Point C, optimum with low price of potatoes, is lower than point E, optimum with high price of potatoes. Callouts describing the arrow marking the movement from AB to AD and the arrow marking the distance between the perpendiculars from points C and E to the vertical axis state, "An increase in the price of potatoes rotates the budget constraint inward … which increases potato consumption if potatoes are a Giffen good."

Why does the consumer respond in this strange way? In this example, meat is a normal good, but potatoes are a strongly inferior good; that is, potatoes are a good that a person buys a lot less of when her income rises and a lot more of when her income falls. In Figure 13, the increase in the price of potatoes makes the consumer poorer in the sense that she moves to a lower indifference curve. Because she is poorer, the income effect makes her want to buy less meat (the normal good) and more potatoes (the inferior good). At the same time, because potatoes have become more expensive relative to meat, the substitution effect makes the consumer want to buy more meat and fewer potatoes. Note that the income and substitution effects push in opposite directions. If the income effect is larger than the substitution effect, as it is in this example, the consumer responds to the higher price of potatoes by buying less meat and more potatoes.

Economists use the term
Giffen good (a good for which an increase in the price raises the quantity demanded) to describe a good that violates the law of demand. (The term is named for economist Robert Giffen, who first noted this possibility.) In this example, potatoes are a Giffen good. Giffen goods are inferior goods for which the income effect dominates the substitution effect. Therefore, they have demand curves that slope upward.

Case Study:The Search for Giffen Goods

Have any actual Giffen goods ever been observed? Some historians suggest that potatoes were a Giffen good during the Irish potato famine of the 19th century. Potatoes were such a large part of people's diet that when the price of potatoes rose, the change had a large income effect. People responded to their reduced living standard by cutting back on the luxury of meat and buying more of the staple food of potatoes. Thus, it is argued that a higher price of potatoes actually raised the quantity of potatoes demanded.

A study by Robert Jensen and Nolan Miller, published in the *American Economic Review* in 2008, produced similar but more concrete evidence for the existence of Giffen goods. These two economists conducted a field experiment for five months in the Chinese province of Hunan. They gave randomly selected households vouchers that subsidized the purchase of rice, a staple in local diets, and used surveys to measure how consumption of rice responded to changes in the price. They found strong evidence that poor households exhibited Giffen behavior. Lowering the price of rice with the subsidy voucher caused households to reduce their consumption of rice, and removing the subsidy had the opposite effect. Jensen and Miller wrote, "To the best of our knowledge, this is the first rigorous empirical evidence of Giffen behavior."

Thus, the theory of consumer choice allows demand curves to slope upward, and sometimes that strange phenomenon actually occurs. As a result, the law of demand we first saw in Chapter 4 is not completely reliable. It is safe to say, however, that Giffen goods are very rare.